Machine Learning

ECE 4370 / ECE 5370

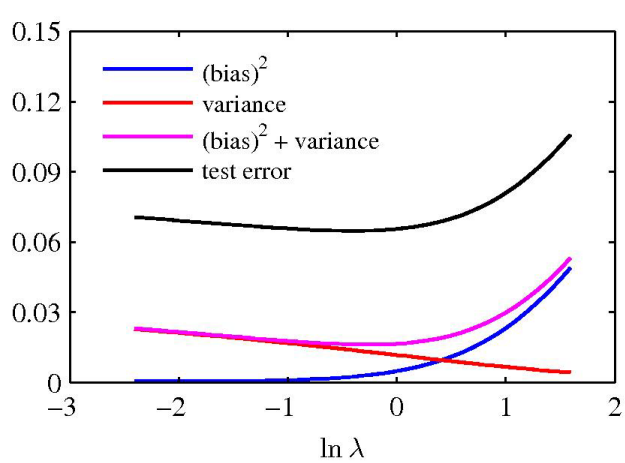
Project 3

1. Generate the data set $\mathcal{D}$ as follows:
   a. $L = 100$
   b. $N = 25$
   c. X contains samples from a uniform distribution $\mathcal{U}(0,1)$.
   d. $t = \sin(2\pi X) + \varepsilon$, where ε contains samples from a Gaussian distribution $\mathcal{N}(0, \sigma = 0.3)$.
2. Select a set of permissible values for the regularization parameter λ.
3. For each value of λ, use the method of "linear regression with non-linear models" to fit Gaussian basis functions to each of the datasets. Use $s = 0.1$.
4. Produce the plot as shown below, where

$$\bar{f}(x) = \frac{1}{L}\sum_{l=1}^{L} f^{(l)}(x)$$

$$(bias)^2 = \frac{1}{N}\sum_{n=1}^{N}\left[\bar{f}\left(x^{(n)}\right) - h\left(x^{(n)}\right)\right]^2$$

$$variance = \frac{1}{N}\sum_{n=1}^{N}\frac{1}{L}\sum_{l=1}^{L}\left[f^{(l)}\left(x^{(n)}\right) - \bar{f}\left(x^{(n)}\right)\right]^2$$

5. The test error curve is the average error for a test data set of 1000 points.

Upload your .m or .py file to Blackboard prior to the deadline.
Name your files as Lastname_Firstname_Project3.*